1 Representation Surgery — Affine Steering (Lec 54)

LMs train to produce **contextual representations** (vectors) to predict the next token. Can we treat these as well-behaved vector spaces for **de-biasing & controlling generation**?

Guardedness: an attribute Z is **guarded** if we **can't classify along it** from the representations (e.g. can't tell a noun's gender from its vector).

Guarded attribute (Q1): the attribute **can only be classified by specific linear probes** — it's protected against general linear classification.

Affine Concept Erasure = an **affine transformation** (linear map + shift) that guards a particular attribute.

What we want (Q2): make vectors from one distribution **look like** those of another (e.g. toxic → non-toxic) — i.e. **guard an attribute with the smallest possible change**, preserving semantics unrelated to Z.

Contributions: (1) a **theoretical justification** for steering vectors as an optimization problem; (2) a more expressive steering function.

Implication (Q3): Affine Steering is **a no-gradient and cheap way to control generation** (and to de-bias) — no fine-tuning, no gradient descent, no RLHF. It protects attributes (e.g. gender) **without damaging** unrelated multi-class classification, and controls generation (e.g. lowers maximum toxicity) with **no fine-tuning**.

2 Graph Explainability — the problem (Lec 53)

Graph data appears in many **high-risk** applications (molecules, transaction networks), so **GNN outputs must be interpretable**. **Graph Explainers** align GNN outputs to our expectations — important in **toxicity detection of molecules, drug prediction, fraud detection** (Q8). E.g. explaining benzene detection = highlighting the correct green ring.

Graph limitation (Q5): graphs can only model **pairwise** relationships between nodes. So a standard GNN explainer **cannot capture specific multi-node group-wise interactions** (like a whole functional group of atoms acting together) — which is why it fails on molecule toxicity.

3 Cell Complexes

A **cell complex** is a **higher-order generalisation of a graph** that **captures interactions between multiple nodes** (not just pairs). The lifting:

- **0-cells** = nodes / vertices
- **1-cells** = edges (augmented nodes connected to their two original nodes)
- **2-cells** = **cycles** (augmented nodes connected to all edges forming that cycle)

Q7: cycles are represented as **2-cells**. A 0-cell = vertex, a 1-cell = edge in traditional graphs (Q10).

Cost (Q4): a conventional cell complex **scales QUADRATICALLY** as the graph size increases — computationally expensive.

Face/boundary (Q10): a p-cell $e^{(p)}$ is a **face/boundary** of a (p+1)-cell $e^{(p+1)}$ if the set of points composing $e^{(p)}$ is a **subset** of those composing $e^{(p+1)}$.

4 FORGE (Framework For Higher-Order Representations in Graph Explanations)

FORGE leverages cell complexes to produce **better, more faithful** graph explanations when group-wise interactions matter. Pipeline:

- **Lift** the graph to a cell complex
- Run the complex through a GNN
- A graph explainer explains the complex
- **Propagate importance** back to the graph (reverse message-passing) → a graph explanation

How FORGE gets linear scale (Q6): it **strategically removes specific types of boundary relations**, bringing the quadratic cost down to **linear** — while keeping the useful higher-order structure.

Who benefits most (Q9): perturbation-based explainer methods benefit the most from FORGE (e.g. FORGE + GNNExplainer is best on real-world data; FORGE + SubgraphX on synthetic). After FORGE, all explainers beat the Random baseline.

Assignment quick-key: Q1 guarded = only specific linear probes classify it; Q2 guard with minimal change to unrelated semantics; Q3 affine steering = no-gradient, cheap generation control; Q4 quadratic scaling; Q5 can't capture multi-node group interactions; Q6 remove boundary relations → linear; Q7 cycles = 2-cells; Q8 graph explainers = molecule toxicity + drug prediction + fraud detection; Q9 perturbation-based methods; Q10 0-cell=vertex & p-cell is a face if its points are a subset of the (p+1)-cell's.

5 Takeaways

- **Affine steering** = cheap, no-gradient representation surgery to erase/guard a concept while preserving unrelated semantics
- **Cell complexes** model group-wise (multi-node) interactions a plain graph can't — 0/1/2-cells = vertices/edges/cycles
- **Conventional complexes scale quadratically; FORGE** drops this to **linear** by removing boundary relations
- **Explainers are task-dependent** — test several before choosing; perturbation-based ones gain most from FORGE